

WLM Foundation Series

Paper 1

Symmetry–Offset Series

The Stability and Uniqueness of 0: Symmetry Interval, Minimal Offset, and the Necessity of ε

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Abstract

We develop a minimal symmetry framework S_0 from four axioms: S_0 -LocalSym, S_0 -Break, V-WF, and totality of generative relations. Main results: 0 is unique perfect symmetry state; all symmetric candidates collapse to 0; ε is unique minimal offset, fully emergent; ε cannot be skipped. Section 4.3 bridges to Paper 2 via the latent directionality of ε .

v2.1 changes from v2: (a) Definition alignment note added to Definition 0.5. (b) Lemma 4.3 strengthened with explicit D_0 reference. (c) Collapse Modes Table enhanced with Paper 0 Reference column.

I_0 (symmetry phase) $\rightarrow \varepsilon$ (unique breaking event) $\rightarrow$ offset chain $\rightarrow \pm\varepsilon$ duality (Paper 2)

0. Framework S_0

0.1 State Space and Observational Equivalence

Let X be a nonempty set of states with $0 \in X$. Let O be a family of observations $O: X \rightarrow Y_O$. Observational equivalence: $x \sim_o y \Leftrightarrow \forall O \in O, O(x) = O(y)$.

Definition 0.1 (Symmetry Interval). $I_0 := [0] \sim_o = \{x \in X : x \sim_o 0\}$.

0.2 Generative Relations

Assumption (Totality): Every $G \in \mathcal{G}$ is a total function $G: X \rightarrow X$.

Definition 0.2 (I_0 -Preserving Generators). $\mathcal{G}_{-}\{I_0\} := \{G \in \mathcal{G} : G(I_0) \subseteq I_0\}$.

Definition 0.3 (Generative Chain). A generative chain is a sequence $x_0 = 0, x_1, x_2, \dots$ with $x_{n+1} = G_n(x_n)$ for some $G_n \in \mathcal{G}$. (Identical to Paper 0, Definition 0.3.)

0.3 Axioms

Axiom S0-LocalSym (Local Symmetry Preservation). For all $x, y \in I_0$ and $G \in \mathcal{G}_{-}\{I_0\}$: $x \sim_0 y$ implies $G(x) \sim_0 G(y)$.

Axiom S0-Break (Unique Symmetry-Breaking). (a) **Existence:** $\exists G \in \mathcal{G} \setminus \mathcal{G}_{-}\{I_0\}$ with $G(0) \not\sim_0 0$. (b) **Minimality by valuation:** the minimal element of $\{v(G(0)) : G(0) \not\sim_0 0\}$ is $v(\varepsilon)$. ε is fully emergent.

Axiom V-WF (Well-Founded Partial Order). V is a well-founded partially ordered set with minimal element $v(0)$. Every nonempty subset of $\{v(x) : v(x) > v(0)\}$ has a minimal element.

Definition 0.4 (Distinguishability Valuation). $v: X \rightarrow (V, \leq)$ satisfies: $v(x) = v(y)$ iff $x \sim_0 y$; $v(0)$ minimal; $x \not\sim_0 0$ implies $v(x) > v(0)$.

Definition 0.5 (Minimal Offset). $\varepsilon \in X$ is a minimal offset if $v(\varepsilon) > v(0)$ and no x satisfies $v(0) < v(x) < v(\varepsilon)$. This definition is identical to the minimal-offset definition in Paper 0 v3.1, where the distinguishability measure $\|x - 0\|$ is realized by $v(\cdot)$ via: $v(x) > v(0) \Leftrightarrow \|x - 0\| > 0$.

Remark (Consistency of Axioms). S0-LocalSym constrains only $\mathcal{G}_{-}\{I_0\}$ within I_0 . S0-Break(a) asserts $G \in \mathcal{G} \setminus \mathcal{G}_{-}\{I_0\}$, outside S0-LocalSym's scope. V-WF is independent of both. The axioms are mutually consistent.

1. The Symmetry Interval

Definition 1.1 (Perfect Symmetry State). $x \in X$ is a perfect symmetry state if $\forall y \in X, x \sim_o y$. Let $Sym_p^{erf} = \{x \in X : \forall y, x \sim_o y\}$.

Theorem 1.1 (Uniqueness). If $x, x' \in Sym_p^{erf}$, then $x \sim_o x'$.

Proof. By definition, $x \sim_o y$ for all y ; in particular $x \sim_o x'$. $\square$

Theorem 1.2 ($Sym_p^{erf} = I_0$). $Sym_p^{erf} = I_0$.

Proof. All perfect symmetry states form one equivalence class $[0]_{\sim_o} = I_0$. $\square$

2. Collapse of Symmetric Candidates

Definition 2.1 (Symmetric Candidate). $Sym^{cand} = \{x \in X : \forall G \in \mathcal{G}_-\{I_0\}, G(x) \sim_o x\}$. Defined relative to $\mathcal{G}_-\{I_0\}$ to exclude escape generators.

Lemma 2.1 (Local Fixity for I_0 -Preserving Generators). If $x \in I_0$ and $G \in \mathcal{G}_-\{I_0\}$, then $G(x) \sim_o x$.

Proof. $x \sim_o 0$; by S0-LocalSym: $G(x) \sim_o G(0)$; since $G(0) \in I_0$: $G(0) \sim_o 0$; by transitivity: $G(x) \sim_o x$. $\square$

Lemma 2.2 (I_0 -Preserving Generators Cannot Escape). If $G(I_0) \subseteq I_0$ and G is total, then $G(x) \in I_0$ for all $x \in I_0$.

Proof. Immediate from definition of $\mathcal{G}_-\{I_0\}$ and totality. $\square$

Theorem 2.1 ($Sym^{cand} = I_0$). $x \in Sym^{cand}$ iff $x \in I_0$.

Proof ($\supseteq$): $x \in Sym^{cand} \Rightarrow G(x) \sim_o x$; since $0 \sim_o G(x)$: $0 \sim_o x$, so $x \in I_0$. Proof ($\subseteq$): Lemma 2.1. $\square$

Theorem 2.2 (Collapse Under Composition). If $G \in \mathcal{G}_-\{I_0\}$ maps Sym^{cand} into itself, then $G(x) \sim_o 0$ for all $x \in Sym^{cand}$.

Proof. Since G maps Sym^{cand} into itself, it preserves I_0 . By Theorem 2.1, $G(x) \in Sym^{cand}$ implies $G(x) \sim_o 0$. $\square$

3. The Emergence and Necessity of ε

Remark (Symmetry Breaking Event). No $G \in \mathcal{G}_{\{l_0\}}$ can create distinguishability within l_0 (S0-LocalSym). The emergence of ε is the unique symmetry-breaking event. By S0-Break(b), ε is fully determined by the generative structure.

Theorem 3.1 (No Universe Without ε). If $\text{MinOffset} = \emptyset$, then $\forall x \in X, x \sim_0 0$.

Proof. Suppose $x \not\sim_0 0$; then $\{v(y) : v(y) > v(0)\}$ is nonempty; by V-WF, it has a minimal element — contradiction. $\square$

Theorem 3.2 (Minimal Offset Uniqueness). If $\varepsilon_1, \varepsilon_2 \in \text{MinOffset}$, then $\varepsilon_1 \sim_0 \varepsilon_2$.

Proof. Both are minimal elements of $\{v(x) : v(x) > v(0)\}$; by V-WF, $v(\varepsilon_1) = v(\varepsilon_2)$, hence $\varepsilon_1 \sim_0 \varepsilon_2$. $\square$

Theorem 3.3 (ε Cannot Be Skipped). If a generative chain contains $x_k \not\sim_0 0$, then $\exists j < k$ with $x_j \sim_0 \varepsilon$.

Proof. $T = \{x_j : v(x_j) > v(0)\}$ is nonempty; by V-WF, has minimal element; that element $\sim_0 \varepsilon$ by Theorem 3.2. $\square$

Theorem 3.4 (Uniqueness Under Weak Well-Foundedness). Even if V is only well-founded above $v(0)$, ε is unique up to observational equivalence.

Proof. $\{v(x) : v(x) > v(0)\}$ is well-founded by V-WF. Any two minimal valuations must be equal; equal valuation implies observational equivalence. $\square$

4. Collapse Modes and Bridge to Paper 2

Theorem 4.1 (Symmetry Collapse). Any chain with no $x_j \sim_0 \varepsilon$ satisfies $x_k \sim_0 0$ for all k .

Proof. Contrapositive of Theorem 3.3. $\square$

Theorem 4.2 (Offset Degeneracy Collapse). If a chain reaches x with $v(x) > v(\varepsilon)$, then some $x_j \sim_0 \varepsilon$ appears earlier.

Proof. $v(x) > v(\varepsilon) > v(0)$ implies $x \not\sim_o 0$; by Theorem 3.3, $x_j \sim_o \varepsilon$ with $j < k$. $\square$

4.3 Bridge to Paper 2: Latent Directionality of ε

Lemma 4.3 (Latent Directionality of ε). ε is the unique state with minimal nonzero valuation. This latent orientation is the structural precursor of the $\pm\varepsilon$ duality developed in Paper 2. Together with Axiom D0 of Paper 0, this latent orientation forces the explicit dual pair $\{+\varepsilon, -\varepsilon\}$ (see Paper 0 Theorem 3).

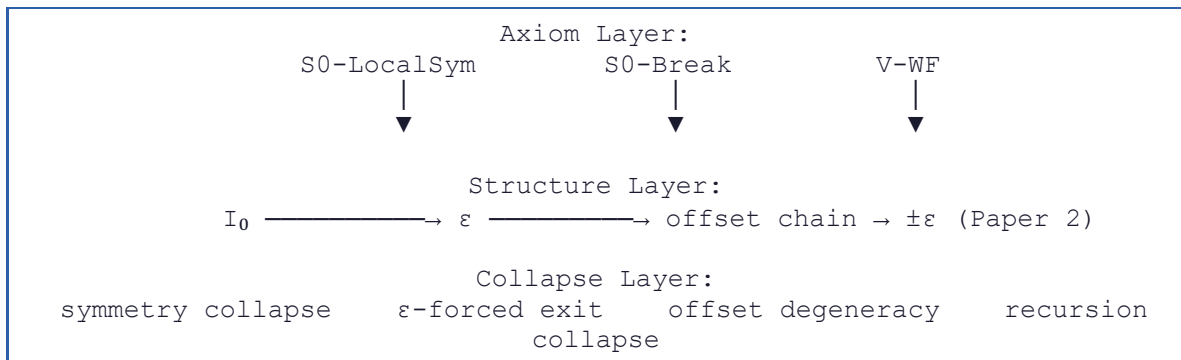
Proof. By Theorem 3.2, ε is unique up to observational equivalence with minimal nonzero valuation. Uniqueness of the minimal nonzero element implies the system has identified one canonical direction of departure from I_0 . Axiom D0 (Paper 0) then forces the dual direction $-\varepsilon$. The explicit construction is in Paper 0 Theorem 3; Paper 1 establishes the necessary precondition. $\square$

Remark (Paper 1 $\rightarrow$ Paper 2 Transition). Paper 1 establishes: ε exists (Thm 3.1), is unique (Thm 3.2), cannot be skipped (Thm 3.3), and carries latent directionality (Lemma 4.3). Paper 0 Theorem 3 shows that once ε exists, Axiom D0 forces $\{+\varepsilon, -\varepsilon\}$. Paper 2 formalizes the $\pm\varepsilon$ duality and the recursion it forces.

5. Structural Summary

S0-LocalSym governs I_0 interior. S0-Break governs unique exit. V-WF ensures ε exists and is minimal. Totality ensures well-posedness. Lemma 4.3 latent directionality $\rightarrow \pm\varepsilon$ duality in Paper 2.

Axiom Interaction Diagram



Collapse Modes Table

Collapse Mode	Trigger Condition	Result	Paper 1 Theorem	Paper 0 Reference
Symmetry collapse	MinOffset = $\emptyset$	All states $\sim_o 0$	Thm 3.1 / 4.1	P0 Theorem 1 (0 is fixed)
ϵ -forced exit failure	Chain skips ϵ	Contradiction (impossible)	Thm 3.3	P0 Theorem 2a (ϵ must exist)
Offset degeneracy	$v(x) > v(\epsilon)$ without ϵ first	ϵ must have appeared earlier	Thm 4.2	P0 Lemma 2b (minimality)
Recursion collapse	No $\pm\epsilon$ directionality (R2 violated)	Fixed point or I_0	Lemma 4.3 (latent dir.)	P0 Theorem 4 + Lemma 4c

Paper 2 will formalize the $\pm\epsilon$ duality and recursion. Paper 1 provides the foundation: 0 is stable, ϵ is necessary and unique, and no generative chain bypasses it.

References

WLM Foundation Series:

- Gu, W. (2026a). The Generative Beginning of a Universe. WLM Foundation Series, Paper 0. ResearchGate.
- Gu, W. (2026b). Wujie Generation Stratum (WGS): The Dual-Layer World Model. ResearchGate. <https://www.researchgate.net/publication/403824388>
- Gu, W. (2026c). The 0-27-0 Generative Cycle. ResearchGate. <https://www.researchgate.net/publication/403884275>

External works:

- Birkhoff, G. (1940). Lattice Theory. American Mathematical Society. — V-WF stated using Birkhoff's formulation of well-founded partial orders.
- Kunen, K. (1980). Set Theory. North-Holland. — Well-foundedness and minimal elements; Theorems 3.1–3.4.
- Mac Lane, S. (1971). Categories for the Working Mathematician. Springer. — $\mathcal{G}_{\{I_0\}}$ and $[0] \sim_o$ have categorical readings.
- Weyl, H. (1952). Symmetry. Princeton University Press. — Symmetry as observational equivalence.

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